

1.

$$\textcircled{1} \sum_{i=0}^4 l_i(x) = 1$$

$$\textcircled{2} \sum_{i=0}^4 f(x_i) l_i(x) = L_4(x)$$

因为 $f(x)$ 是 4 次多项式, 且多项式插值唯一

$$\Rightarrow \sum_{i=0}^4 f(x_i) l_i(x) = f(x)$$

$$\textcircled{3} f[x_0, x_1, x_2, x_3, x_4] = \frac{f^{(4)}(\frac{1}{2})}{4!} = 30$$

$$\begin{aligned}
 2. \quad L_3(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} \cdot y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \cdot y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} \cdot y_2 \\
 &\quad + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} \cdot y_3 = -2x^3 + \frac{16}{3}x^2 - \frac{10}{3}x + 3
 \end{aligned}$$

$$\begin{aligned}
 N_3(x) &= f(x_0) + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) \\
 &\quad + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2)
 \end{aligned}$$

$$\Rightarrow f[x_0, x_1] = 0$$

$$f[x_0, x_1, x_2] = \frac{f[x_0, x_2] - f[x_0, x_1]}{x_2 - x_1} = \frac{1}{3}$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_0, x_1, x_3] - f[x_0, x_1, x_2]}{x_3 - x_2} = -2$$

$$\Rightarrow N_3(x) = -2x^3 + \frac{16}{3}x^2 - \frac{10}{3}x + 3$$

3. 步长为0.1, 求插值余项

$$R_{10}(x) = f(x) - L_{10}(x) = \frac{f^{(11)}(\xi)}{11!} w_{11}(x)$$

$$\Rightarrow w_{11}(x) = \prod_{i=0}^n (x - x_i)$$

$$f^{(11)}(\xi) = e^{\xi-2}, \text{ 在 } [0, 1] \text{ 区间上, } \max |f^{(11)}(\xi)| = e^{-1} \approx 0.3679$$

$$\text{则 } \max (\prod_{i=0}^{10} (x - x_i)) \approx 4.2 \times 10^{-6}$$

$$\Rightarrow |R_{10}(x)| \leq \frac{0.3679}{11!} \times 4.2 \times 10^{-6} \approx 4 \times 10^{-14}$$

插值余项估计为 4×10^{-14}

$$4. |f(x) - P(x)| \leq \frac{M_2}{8} h^2$$

$f''(x) = e^x$, 在区间 $[0, 1]$, $|f''(x)| = e^x$ 的 $x=1$ 处, 所以 $M_2 = e' = e$

$$h^2 = 0.01$$

$$|f(x) - P(x)| \leq \frac{e}{8} \times 0.01 \doteq 0.339785 \times 0.01 \doteq 0.0034$$

$\Rightarrow$ 插值误差上界约为 0.0034

5. 三点三次 Hermite 插值

商差表 1

x_k	$f(x_k)$	$f[x_k, x_{k+1}]$	$f[x_k, x_{k+1}, x_{k+2}]$	$f[x_k, x_{k+1}, x_{k+2}, x_{k+3}]$
0	0			
0.5	1	2		
0.5	1	2	0	
1	1	0	-4	-4

$$H_3(x) = f(x_0) + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) \\ + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2) = -4x^3 + 4x^2 + x$$

商差表 2

x_k	$f(x_k)$	$f[x_k, x_{k+1}]$	$f[x_k, x_{k+1}, x_{k+2}]$	$f[x_k, x_{k+1}, x_{k+2}, x_{k+3}]$	$f[x_k, \dots, x_{k+4}]$
0	0				
0.5	1	2			
0.5	1	2	0		
1	1	0	-4	4	
1	1	-3	-6	-4	-8

$$H_4(x) = -4x^3 + 4x^2 + x$$

6. 设 $P_2(x) = C_0 + C_1x + C_2x^2$, $\mathbb{R} | \varphi_0 = 1, \varphi_1 = x, \varphi_2 = x^2$

法方程组

$$\begin{pmatrix} (\varphi_0, \varphi_0) & (\varphi_1, \varphi_0) & (\varphi_2, \varphi_0) \\ (\varphi_0, \varphi_1) & (\varphi_1, \varphi_1) & (\varphi_2, \varphi_1) \\ (\varphi_0, \varphi_2) & (\varphi_1, \varphi_2) & (\varphi_2, \varphi_2) \end{pmatrix} \begin{pmatrix} C_0 \\ C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} (f, \varphi_0) \\ (f, \varphi_1) \\ (f, \varphi_2) \end{pmatrix}$$

$$\Rightarrow (\varphi_0, \varphi_0) = \int_0^\pi 1 \cdot 1 dx = \pi$$

$$(\varphi_0, \varphi_1) = (\varphi_1, \varphi_0) = \int_0^\pi 1 \cdot x dx = \frac{1}{2}\pi^2$$

$$(\varphi_0, \varphi_2) = (\varphi_2, \varphi_0) = (\varphi_1, \varphi_1) = \int_0^\pi 1 \cdot x^2 dx = \frac{1}{3}\pi^3$$

$$(\varphi_1, \varphi_2) = (\varphi_2, \varphi_1) = \int_0^\pi x \cdot x^2 dx = \frac{1}{4}\pi^4$$

$$(\varphi_2, \varphi_2) = \int_0^\pi x^2 \cdot x^2 dx = \frac{1}{5}\pi^5$$

$$(f, \varphi_0) = \int_0^\pi f(x) dx = \frac{\pi^3}{3} - 2$$

$$(f, \varphi_1) = \int_0^\pi x f(x) dx = \frac{\pi^4}{4} - 2\pi$$

$$(f, \varphi_2) = \int_0^\pi x^2 f(x) dx = \frac{\pi^5}{5} - 3\pi^2 + 12$$

$$\text{解得} \begin{cases} C_0 = -\frac{36}{\pi} + \frac{360}{\pi^3} \doteq 0.157 \\ C_1 = \frac{228}{\pi^2} - \frac{2160}{\pi^4} \doteq 0.917 \\ C_2 = 1 - \frac{240}{\pi^3} + \frac{2160}{\pi^5} \doteq 0.321 \end{cases}$$

$$\text{得 } P(x) = 0.157 + 0.917x + 0.321x^2$$

7. 设 $p(x) = x^3 + ax^2 + bx + c$, 即使 $E = \int_{-1}^1 p(x)^2 dx$ 最小

$$\frac{\partial E}{\partial a} = \frac{4}{5}a + \frac{4}{3}c = 0$$

$$\frac{\partial E}{\partial b} = \frac{4}{5} + \frac{4}{3}b = 0$$

$$\frac{\partial E}{\partial c} = \frac{4}{3}a + 4c = 0$$

$$\left. \begin{array}{l} \frac{\partial E}{\partial a} = \frac{4}{5}a + \frac{4}{3}c = 0 \\ \frac{\partial E}{\partial b} = \frac{4}{5} + \frac{4}{3}b = 0 \\ \frac{\partial E}{\partial c} = \frac{4}{3}a + 4c = 0 \end{array} \right\} \rightarrow \text{解得} \begin{cases} a = 0 \\ b = -\frac{3}{5} \\ c = 0 \end{cases}$$

$$\Rightarrow p(x) = x^3 - \frac{3}{5}x$$

$$8. A = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$r(A) = 2, r(A, b) = 3$$

由 $r(A) < r(A, b)$, $AX = b$ 无解

$$(A^T A, A^T b) = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 5 & 0 & 3 \\ 2 & 0 & 4 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & \frac{5}{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

由此可得 $x = c \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ \frac{5}{3} \\ 0 \end{pmatrix}$, 其中 c 为任意常数

$$\text{取 } x = \begin{pmatrix} 1 \\ \frac{5}{3} \\ 0 \end{pmatrix}$$

9. 使用最小二乘拟合法

$$\varphi(0) = (1, 1, 1, 1, 1)^T$$

$$\varphi(1) = (-2, -1, 0, 1, 2)^T$$

$$\varphi(2) = (4, 1, 0, 1, 4)^T$$

$$f = (0, 1, 2, 1, 0)^T$$

$$\Rightarrow \begin{pmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & (\varphi_0, \varphi_2) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & (\varphi_1, \varphi_2) \\ (\varphi_2, \varphi_0) & (\varphi_2, \varphi_1) & (\varphi_2, \varphi_2) \end{pmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} (f, \varphi_0) \\ (f, \varphi_1) \\ (f, \varphi_2) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} \Rightarrow a = -\frac{3}{7}, b = 0, c = \frac{58}{35}$$

$$\Rightarrow p(x) = -\frac{3}{7}x^2 + \frac{58}{35}$$

$$10. \text{ i\ddot{z}} y = a \varphi_1(x) + b \varphi_2(x), \quad \varphi_1(x) = x, \quad \varphi_2(x) = \sin^2 \frac{\pi x}{b}$$

$$\Rightarrow A = \begin{pmatrix} \varphi_1(x_1) & \varphi_2(x_1) \\ \vdots & \vdots \\ \varphi_1(x_4) & \varphi_2(x_4) \end{pmatrix} = \begin{pmatrix} 1 & 1/4 \\ 2 & 3/4 \\ 3 & 1 \\ 4 & 3/4 \end{pmatrix}$$

$$\vec{x} = \begin{pmatrix} a \\ b \end{pmatrix}, \quad \vec{b} = (2, 1, 0, 1)^T$$

$$A \vec{x} = \vec{b}$$

$$\rightarrow \text{BP } \vec{x} = (A^H A)^{-1} A^H \vec{b}$$

$$= \left(\frac{32}{89}, -\frac{32}{89} \right)^T$$

$$\Rightarrow y = \frac{32}{89} x - \frac{32}{89} \sin^2 \frac{\pi x}{b}$$

11. 两边取对数可得 $\ln y = \ln a + bx$

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1.25 \\ 1 & 1.5 \\ 1 & 1.75 \\ 1 & 2 \end{pmatrix} \quad \vec{x} = \begin{pmatrix} \ln a \\ b \end{pmatrix}$$

$$\vec{b} = \begin{pmatrix} \ln 5.1 \\ \ln 5.79 \\ \ln 6.53 \\ \ln 7.45 \\ \ln 8.46 \end{pmatrix} \doteq \begin{pmatrix} 1.63 \\ 1.75 \\ 1.88 \\ 2 \\ 2.14 \end{pmatrix}$$

$$\text{求 } A\vec{x} = \vec{b}$$

$$\Rightarrow \vec{x} = (A^H A)^{-1} A^H \vec{b}$$

$$\doteq (1.12, 0.505)^T$$

$$\Rightarrow e^{\ln a} \doteq 3.074$$

$$\Rightarrow y = 3.074 e^{0.505x}$$

$$\text{当 } x=1.35 \text{ 时, } \hat{y} = 6.08$$

12. Simpson 公式:

$$\int_a^b f(x) dx = \frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)] , R_2 = \frac{(b-a)}{-180} (\frac{h}{2})^4 f^{(4)}(\xi)$$

$$\Rightarrow \frac{1}{n} \text{长 } h = \frac{b-a}{n} , x_k = a + kh$$

$$\int_a^b f(x) dx = \sum_{i=0}^{m-1} \int_{x_{2i}}^{x_{2i+2}} f(x) dx$$

在 $[x_{2i}, x_{2i+2}]$ 上用 Simpson 公式

$$\text{累加求和 Simpson 求积} \Rightarrow S_n(f) \doteq \frac{h}{3} [f(a) + 4 \sum_{i=0}^{m-1} f(x_{2i+1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + f(b)]$$

$$\text{代入得 } I \doteq \frac{0.8}{3} [1 - 12 + 2] = -2.4$$

$$13. \text{梯形公式} = \int_a^b f(x) dx = \frac{b-a}{2} [f(a) + f(b)] \quad R_1 \approx \frac{(b-a)^3}{-12} f''(\xi)$$

$$\text{复化梯形公式} \quad \int_a^b f(x) dx = \frac{h}{2} [f(a) + 2 \sum_{k=1}^{n-1} f(x_k) + f(b)]$$

$$\Rightarrow I_1 = \int_1^{2.2} \sin \frac{1}{x} dx = \frac{0.6}{2} [0.85 + 2 \times 0.59 + 0.44] = 0.741$$

$$|R_1| = \left| \frac{1.2 \times 0.36}{-1.2} f''(\xi) \right| = 0.36 \cdot \left| \frac{2}{x^3} \cos \frac{1}{x} - \frac{1}{x^4} \sin \frac{1}{x} \right| \leq 0.36 \times 0.42 = 0.01512$$

$$\Rightarrow I_2 = \frac{0.6}{3} [0.85 + 4 \times 0.59 + 0.44] = 0.2 \times 3.65 = 0.73$$

$$|R_2| = \left| -\frac{1.2 \times 0.1296}{180} f^{(4)}(\xi) \right| = 0.000864 |f^{(4)}(\xi)| \leq 0.000864 \times 23 \leq 0.02$$

$$14. I_1 = \frac{0.2}{2} [1 + 4 - 2 - 6 - 2 + 2 + 6 + 2] = 0.5$$

$$I_2 = \frac{0.2}{3} [1 + 8 - 4 - 6 - 4 + 2 + 12 + 2] = \frac{11}{15} \approx 0.7333$$

15. 有3个求积节点, 即需要 $f(x) = x^{2n-1}$ 精确成立. 即 $f(x) = x^5$ 都成立

$$\Rightarrow \begin{cases} \int_0^2 1 dx = 2 = A_1 + A_2 + A_3 \\ \int_0^2 x dx = 2 = A_2 + A_3 \\ \int_0^2 x^2 dx = \frac{8}{3} = A_2 + 4A_3 \end{cases} \quad \text{求解得} \begin{cases} A_1 = \frac{1}{3} \\ A_2 = \frac{4}{3} \\ A_3 = \frac{1}{3} \end{cases}$$

检验对更高精度是否成立

$$\int_0^2 x^3 dx = 4 = \frac{1}{3} \cdot 0 + \frac{4}{3} \cdot 1 + \frac{1}{3} \cdot 8, \text{ 对 } x^3 \text{ 精确}$$

$$\int_0^2 x^4 dx = \frac{32}{5} \neq \frac{1}{3} \cdot 0 + \frac{4}{3} \cdot 1 + \frac{1}{3} \cdot 16 = \frac{20}{3}, \text{ 不相等}$$

$\Rightarrow$ 代数精度仅为3, 不满足5, 因此不是 Gauss 型求积公式

16. Gauss - Legendre 求积公式, 区间在 $[-1, 1]$, 先做映射

$$\text{令 } x = \frac{1}{6}(1+t), \text{ 则 } I = \int_{-1}^1 \frac{\sqrt{3}}{3} \cdot \frac{\sqrt{1-t}}{\sqrt{1+t}} dt = \frac{\sqrt{3}}{3} \int_{-1}^1 g(t) dt \Rightarrow g(x) = \frac{\sqrt{1-t}}{\sqrt{1+t}}$$

$$\Rightarrow 3 \text{ 点求积公式为 } I' = \frac{5}{9} g(-\sqrt{\frac{3}{5}}) + \frac{8}{9} g(0) + \frac{5}{9} g(\sqrt{\frac{3}{5}}) = \frac{8}{9} + \frac{10}{9} \sqrt{\frac{5}{2}}$$

$$I = \frac{\sqrt{3}}{3} \times I' = \frac{\sqrt{3}}{27} (18 + 5\sqrt{10}) \doteq 1.5275$$

Gauss - Chebyshev

$$\text{三} \text{ 点 G-C 求积公式为 } \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} g(x) dx \doteq \frac{\pi}{3} \left[g(-\frac{\sqrt{3}}{2}) + g(0) + g(\frac{\sqrt{3}}{2}) \right]$$

$$\text{原式 } I = \frac{\sqrt{3}}{3} \int_{-1}^1 \frac{1+t}{\sqrt{1-t^2}} dt \Rightarrow g(x) = 1+t$$

$$\Rightarrow I = \frac{\sqrt{3}}{3} \times \frac{\pi}{3} \left(1 - \frac{\sqrt{3}}{2} + 1 + 1 + \frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}}{3} \pi \doteq 1.8138$$

17. 改进的 Euler 格式为

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_n + hf(x_n, y_n))]$$

$$\Rightarrow y_{n+1} = y_n + \frac{0.1}{2} [x_n + y_n^2 + x_{n+1} + [y_n + 0.1 \times (x_n + y_n^2)]^2]$$

$$\text{取 } y_1 = 1 + \frac{1}{20} [1 + 0.1 + (1.1)^2] = 1.1155$$

$$y_2 = 1.1155 + 0.05 \times [0.2 + 1.5623 + 1.3443] = 1.27083$$

$\Rightarrow y(x)$ 在 $x=0.2$ 处的近似值为 1.2708

18. (1) 其为二阶 Runge-Kutta 方法, 其局部截断误差为 $O(h^3)$,
因此整体截断误差为 $O(h^2)$, 即整体精度为二阶

(2) 当 $f(x, y) = \lambda y$ 时

$$y_{n+1} = y_n + h\lambda y_n + \frac{h^2}{4}\lambda^2 y_n$$

令 $z = h\lambda$, 则稳定性函数为 $R(z) = 1 + z + \frac{z^2}{4}$

对于绝对稳定性, 要求 $|R(z)| < 1$.

$\Rightarrow$ 解得当且仅当 $z \in (-4, 0)$ 时, $|R(z)| < 1$ 成立

$$\text{即 } 0 < h < -\frac{4}{\lambda}$$

即算法绝对稳定当 $h \in (0, -\frac{4}{\lambda})$ 时

$$19. (1) A = \begin{pmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -1 & -6 & 4 \\ 1 & 5 & -3 \\ 0 & 2 & -1 \end{pmatrix} \Rightarrow \|A\|_1 = \max(4, 5, 4) = 5, \\ \|A^{-1}\|_1 = \max(2, 13, 8) = 13 \\ \text{cond}_1(A) = \|A\|_1 \|A^{-1}\|_1 = 5 \times 13 = 65$$

$$(2) \delta b = b' - b = (0.001, -0.002, 0)^T \\ \|\delta b\|_1 = 0.003, \quad \|b\|_1 = 3 \quad \Rightarrow \frac{\|\delta b\|_1}{\|b\|_1} = 0.001$$

利用条件数估计解的相对误差

$$\Rightarrow \frac{\|\delta x\|_1}{\|x\|_1} \leq \text{cond}_1(A) \times \frac{\|\delta b\|_1}{\|b\|_1} = 0.065, \quad \text{解的相对误差估计为 } 6.5\%$$

$$(3) \text{Jacobi 格式} \begin{cases} x_1^{(k+1)} = 1 - 2x_2^{(k)} + 2x_3^{(k)} \\ x_2^{(k+1)} = 1 - x_1^{(k)} - x_3^{(k)} \\ x_3^{(k+1)} = 1 - 2x_1^{(k)} - 2x_2^{(k)} \end{cases} \quad \text{Gauss-Seidel 格式} \begin{cases} x_1^{(k+1)} = 1 - 2x_2^{(k)} + 2x_3^{(k)} \\ x_2^{(k+1)} = 1 - x_1^{(k+1)} - x_3^{(k)} \\ x_3^{(k+1)} = 1 - 2x_1^{(k+1)} - 2x_2^{(k+1)} \end{cases}$$

$$(4) M_J = \begin{pmatrix} 0 & 2 & -2 \\ 1 & 0 & 1 \\ 2 & 2 & 0 \end{pmatrix}, \quad \text{收敛的必要条件是迭代矩阵的谱半径 } \rho(M_J) < 1 \\ \text{计算得 } \det(M_J - \lambda I) = -\lambda^3, \text{ 设 } -\lambda^3 = 0, \text{ 得 } \lambda = 0 \text{ (三重根)} \\ \rho(M_J) = 0 < 1, \text{ Jacobi 迭代收敛}$$

20. (1) 迭代格式 $x_{n+1} = e^{-x_n}$ 的迭代函数 $g(x) = e^{-x}$, 其导数为 $g'(x) = -e^{-x}$

设 $x = e^{-x}$ 的解为 α , 其满足 $\alpha = e^{-\alpha}$, $\alpha > 0$

当 $x = \alpha$ 时, 有 $|g'(\alpha)| = e^{-\alpha} < 1$, 满足局部收敛条件

且 $x_0 = 0.5$ 在区间 $[0, 1]$ 内, 迭代值均在该区间内, 故迭代收敛,

(2) 牛顿迭代公式为 $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, 其中 $f(x) = x - e^{-x}$, $f'(x) = 1 + e^{-x}$

$$\Rightarrow x_{n+1} = \frac{e^{-x_n}(x_n + 1)}{1 + e^{-x_n}}, \text{ 取 } x_0 = 0.5$$

$f'(x) > 0$, $f''(x) > 0$, 故 $f(x)$ 是凸函数且递增

$f(0.5) < 0$, $f(0.6) > 0$, 解 α 在 $(0.5, 0.6)$ 内

初始值 $x_0 = 0.5$ 接近解 α , 迭代收敛

21.

(1) 由 $f(x^*) = f'(x^*) = f''(x^*) = 0, f'''(x^*) \neq 0$ 泰勒展开得 $f(x_n) \approx \frac{f'''(x^*)}{6} e_n^3, f'(x_n) \approx \frac{f'''(x^*)}{2} e_n^2$ Newton 迭代公式 $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, 代入得 $e_{n+1} = \frac{2}{3} e_n$ 故收敛性为线性, 收敛因子为 $\frac{2}{3}$ (2) 修改迭代公式为 $x_{n+1} = x_n - 3 \frac{f(x_n)}{f'(x_n)}$ 同样泰勒展开, 得 $e_{n+1} = O(e_n^2)$, 故收敛阶为二阶